Discrete Structures: Basic Structures – P3: Sequences

Shuai Zhang

Department of Computer Science
University of Central Oklahoma

Outline For This Chapter

- Basic Structures
  - Set
  - Function
  - Sequence
    - Packages are scanned in order of arrival
    - We can use index to show the order: 0-based, or 1-based.
  - Summation
    - Each truck's total load stays within the capacity

Definition of Sequences

- From the name of this terminology "sequence", we can get the feeling, this structure is roughly a list with **ordered terms/elements**
- A sequence can be defined by a **function** from a **subset** of the set of integers (usually either the set $\mathbb{N} = \{0,1,2 \cdots\}$ or the set $\mathbb{Z}^+ = \{1,2,3 \cdots\}$) to a set *S*.
- We use the notation $\boldsymbol{a_n}$ to denote the image of the integer n. We call a_n a ***term*** of the sequence. And the sequence is denoted as $\{\boldsymbol{a_n}\}$
- Which part defines the order of the terms?
  - As long as this function is well defined, it's sufficient to generate the entire sequence.

Examples of Sequences 1

- For the sequence $\{a_n\}$, where $a_n = \frac{1}{n}$ and n is a natural number, the first four terms are $1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}$
- The sequence defined by the exponential function $f(x) = ar^x$ is $a, ar, ar^2, \cdots, ar^n, \cdots$
  - This is called ***geometric progression***, with ***initial term*** $\boldsymbol{a}$ and the ***common ratio*** $\boldsymbol{r}$ (both $\boldsymbol{a}$ and $\boldsymbol{r}$ are real numbers)
  - The sequence $\{b_n\}$, where $b_n = (-1)^n$ is a geometric progression, its first four elements are $b_0 = 1, b_1 = -1, b_2 = 1, b_3 = -1$
  - The sequence $\{c_n\}$, where $c_n = 6 \cdot (1/3)^n$ is a geometric progression, its first four elements are $c_0 = 6, c_1 = 2, c_2 = \frac{2}{3}, c_3 = \frac{2}{9}$

Examples of Sequences 2

- The sequence defined by the linear function $f(x) = dx + a$ is $a, a + d, a + 2d, \cdots, a + nd, \cdots$
  - This is called ***arithmetic progression***, with ***initial term*** $\boldsymbol{a}$ and the ***common difference*** $\boldsymbol{d}$ (both $\boldsymbol{a}$ and $\boldsymbol{d}$ are real numbers)
  - The sequence $\{s_n\}$, where $s_n = -1 + 4n$ is an arithmetic progression, and its first four terms are
    $$s_0 = -1, s_1 = 3, s_2 = 7, s_3 = 11$$
  - The sequence $\{t_n\}$, where $t_n = 7 - 3n$ is an arithmetic progression, and its first four terms are
    $$t_0 = 7, t_1 = 4, t_2 = 1, t_3 = -2$$

Represent a Sequence

- Roster method
  - Specify the first several terms with an easy pattern
    - E.g. $1, 3, 5, 7, 9, \ldots$
- A formula
  - E.g. $s_n = -1 + 4n$ for sequence $\{s_n\}$
- Recurrence relation
  - Terms can be specified by previous terms
  - **Initial terms are needed**
  - E.g. for geometric progression, $a_{n+1} = a_n \cdot r$
    - If a_0 is given, then the entire sequence can be defined
  - E.g. for arithmetic progression, $a_{n+1} = a_n + d$

Recurrence Relation

- Definition: A ***recurrence relation*** for the sequence $\{a_n\}$ is an equation that expresses a_n in terms of one or more of the previous terms of the sequence, namely, $a_0, a_1, \ldots, a_{n-1}$, for all integers n with $n \geq n_0$ where n_0 is a nonnegative integer.
- A sequence is called a ***solution*** of a recurrence relation if its terms satisfy the recurrence relation.
  - With different initial terms, we will have different solutions to the recurrence relation
  - Recurrence relation + initial terms = **closed formula**
    - Or just simply call it a formula for this sequence

Examples of Recurrence Relations

- The ***Fibonacci sequence***, $f_0, f_1, f_2, \ldots$, is defined by the initial conditions $f_0 = 0, f_1 = 1$, and the recurrence relation $f_n = f_{n-1} + f_{n-2}$ for $n = 2, 3, 4, \ldots$.
  - $f_2 = f_1 + f_0 = 1 + 0 = 1$
  - $f_3 = f_2 + f_1 = 1 + 1 = 2$
  - $f_4 = f_3 + f_2 = 2 + 1 = 3$
  - $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, \ldots$
- The ***Lucas sequence***, $L_1, L_2, L_3, \ldots$, is defined by the initial conditions $L_1 = 1, L_2 = 3$, and the recurrence relation $L_n = L_{n-1} + L_{n-2}$ for $n = 3, 4, 5, \ldots$.
  - $1, 3, 4, 7, 11, 18, 29, 47, 76, 123, \ldots$

Get a Formula from Recurrence Relation 1

- Let $\{a_n\}$ be a sequence that satisfies the recurrence relation $a_n = a_{n-1} + 3$ for $n = 1,2,3, \ldots$, and $a_1 = 2$. Find a formula for this sequence
  - In a forward way:
    - $a_2 = 2 + 3$
    - $a_3 = (2 + 3) + 3 = 2 + 3 \cdot 2$
    - …..
    - $a_n = a_{n-1} + 3 = \big(2 + 3 \cdot (n-2)\big) + 3 = 2 + 3(n-1)$
  - In a backward way:

$$\begin{aligned} a_n &= a_{n-1} + 3 \\ &= (a_{n-2}+3) + 3 = a_{n-2} + 3 \cdot 2 \\ &= (a_{n-3}+3) + 3 \cdot 2 = a_{n-3} + 3 \cdot 3 \\ &\vdots \\ &= a_2 + 3(n-2) = (a_1 + 3) + 3(n-2) = 2 + 3(n-1) \end{aligned}$$

The technique we used here is call **iteration**: check the terms one-by-one.

Get a Formula from Recurrence Relation 2

- Suppose that a person deposits \$10,000 is a savings account at a bank yielding 11% per year with interest compounded annually. How much will be in the account after 30 years?
  - Let P_n denote the amount in the account after n years
  - $P_n = P_{n-1} + 0.11P_{n-1} = (1.11)P_{n-1}$
  - $P_1 = (1.11)P_0$
  - $P_2 = (1.11)P_1 = (1.11)^2 P_0$
  - $P_3 = (1.11)P_2 = (1.11)^3 P_0$
  - $\vdots$
  - $P_n = (1.11)P_{n-1} = (1.11)^n P_0$
  - So $P_{30} = (1.11)^{30} 10{,}000 = \$228{,}922.97$

Get Recurrence Relation from a Formula

- Suppose that $\{a_n\}$ is a sequence of integers defined by $a_n = n!$, the value of the factorial function at the integer n, where $n = 1, 2, 3, \ldots$
  - $n! = n(n-1)(n-2) \ldots 2 \cdot 1 = n(n-1)! = na_{n-1}$

- Give a recurrence relation of the sequence $\{a_n\}$, $n = 1,2,3, \ldots$ if $a_n = n(n+1)$
  - $$\begin{aligned} a_n &= n(n+1) \\ &= ((n-1)+1)(((n-1)+1)+1) \\ &= (n-1)\big((n-1)+1\big) + (n-1) + \big((n-1)+1\big) + 1 \\ &= a_{n-1} + 2n \end{aligned}$$
  - The key is to construct previous terms

Summation

- Summation notation is used to express the sum of a sequence $\{a_n\}$ and any one of the following forms may be used.

$$\sum_{j=m}^{n} a_j = a_m + a_{m+1} + \cdots + a_n$$

$$\sum_{j=m}^{n} a_j = a_m + a_{m+1} + \cdots + a_n$$

$$\sum_{m \le j \le n} a_j = a_m + a_{m+1} + \cdots + a_n$$

  - The index of the summation runs through all integers starting with its **lower limit** *m* and ending with its **upper limit** *n*.

Some Useful Summation Formulas

Sum	Closed Form
$\sum_{k=0}^{n} ar^k (r \neq 0)$	$\frac{ar^{n+1} - a}{r - 1}, (r \neq 1)$
$\sum_{k=1}^{n} k$	$\frac{n(n+1)}{2}$
$\sum_{k=1}^{n} k^2$	$\frac{n(n+1)(2n+1)}{6}$
$\sum_{k=1}^{n} k^3$	$(\frac{n(n+1)}{2})^2$
$\sum_{k=0}^{\infty} x^k, \lvert x \rvert < 1$	$\frac{1}{1-x}$

To be used in the next chapter

Examples of Summations 1

- Use summation notation to express the sum of the first 100 terms of the sequence $\{a_j\}$, where $a_j = 1/j$ for $j = 1, 2, 3, \ldots$

$$\sum_{j=1}^{100} \frac{1}{j}$$

- What is the value of $\sum_{j=1}^{5} j^2$

$$\sum_{j=1}^{5} j^2 = 1^2 + 2^2 + 3^2 + 4^2 + 5^2$$
$$= 1 + 4 + 9 + 16 + 25 = 55$$

Examples of Summations 2

- Find the sequence for the following recurrence relation $a_n = -2 + 3a_{n-1}, a_0 = 3$
  - $a_n = -2 + 3a_{n-1}$

$$= -2 + 3(-2 + 3a_{n-2}) = -2 + 3(-2) + 9a_{n-2}$$
$$= -2 + 3(-2) + 9(-2 + 3a_{n-3})$$
$$= -2 + 3(-2) + 9(-2) + 27a_{n-4}$$

… …

$$= -2(1 + 3 + 9 + \cdots + 3^{n-1}) + 3^n a_{n-n}$$
$$= -2\sum_{k=1}^{n-1} 3^k + 3^n\, a_0$$
$$= -2\frac{3^n - 1}{2} + 3^{n+1} = 1 + 2 * 3^n$$

Examples of Summations 3

- **Nested summation notations**: **remove one by one**

$$\sum_{i=1}^{4}\sum_{j=1}^{3} ij = \sum_{i=1}^{4}(i+2i+3i) = \sum_{i=1}^{4} 6i = 6\sum_{i=1}^{4} i = 6\cdot\frac{4\cdot 5}{2} = 60$$

$$\sum_{i=1}^{m}\sum_{j=1}^{n} i(i+j) = \sum_{i=1}^{m}\sum_{j=1}^{n}(i^2+ij)$$

$$= \sum_{i=1}^{m}\left((i^2+i)+(i^2+2i)+\cdots+(i^2+ni)\right)$$

$$= \sum_{i=1}^{m}(n*i^2+n*\frac{n+1}{2}*i)$$

$$= n*\sum_{i=1}^{m} i^2 + n*\frac{n+1}{2}\sum_{i=1}^{m} i$$

Practice 1

- 1. Let $a_n=2^n+5\cdot 3^n$ for n=0,1,2,…
  - Find a_0, a_1, and a_2.
  - Show that $a_2 = 5a_1 - 6a_0$, and $a_3 = 5a_2 - 6a_1$.
  - Show that $a_{n-2} = 5a_{n-1} - 6a_{n-2}$

Practice 2

- 1. Let $a_n=2^n+5\cdot 3^n$ for n=0,1,2,…
  - Find a_0, a_1, a_2, and a_3.
    - $a_0 =1+5=6$; $a_1 = 2+15=17$; $a_2 = 4+45=49$; $a_3 = 8+135=143$
  - Show that $a_2 = 5a_1 - 6a_0$, and $a_3 = 5a_2 - 6a_1$.
    - $49=5\cdot 17-6\cdot 6=85-36$; $143=5\cdot 49 - 6\cdot 17=245-102$
  - Show that $a_n = 5a_{n-1} - 6a_{n-2}$
    - $a_{n-1} = 2^{n-1}+5\cdot 3^{n-1}$
    - $a_{n-2} = 2^{n-2}+5\cdot 3^{n-2}$
    - $5a_{n-1} - 6a_{n-2} =5\cdot(2^{n-1}+5\cdot 3^{n-1}) – 6\cdot(2^{n-2}+5\cdot 3^{n-2})$
      $= 5\cdot 2^{n-1}+25\cdot 3^{n-1} – 6\cdot 2^{n-2} – 30\cdot 3^{n-2}$
      $= 5\cdot 2^{n-1}+25\cdot 3^{n-1} – 3\cdot 2\cdot 2^{n-2} – 10\cdot 3\cdot 3^{n-2}$
      $= 5\cdot 2^{n-1}+25\cdot 3^{n-1} – 3\cdot 2^{n-1} – 10\cdot 3^{n-1}$
      $= 2\cdot 2^{n-1}+15\cdot 3^{n-1} = 2^n+5\cdot 3\cdot 3^{n-1}$
      $= 2^n+5\cdot 3^n = a_n$

Practice 3

- 2. Give a recurrence relation of the sequence $\{a_n\}$, $n = 1,2,3, \ldots$ if $a_n = 1 + (-1)^n$

Practice 4

- 2. Give a recurrence relation of the sequence $\{a_n\}$, $n = 1,2,3, \ldots$ if $a_n = 1 + (-1)^n$
  - $a_n = a_{n-2}$
  - If you want to make this into a closed formula, you can further add $a_1 = 0$ and $a_2 = 2$.

Practice 5

- 3. Compute each of the double sums.
  - $\sum_{\boldsymbol{i}=\mathbf{0}}^{\mathbf{2}} \sum_{j=0}^{3} (2i + 3j)$
  - $\sum_{m=3}^{5} \sum_{n=2}^{4} (m - n)(m + n)$

Practice 6

- 3. Compute each of the double sums.
  - $\sum_{\boldsymbol{i=0}}^{\mathbf{2}} \sum_{j=0}^{3} (2i + 3j)$
    - $\sum_{\boldsymbol{i=0}}^{\mathbf{2}} \sum_{j=0}^{3} (2i + 3j) = \sum_{i=0}^{2} \big((2i) + (2i + 3) + (2i + 6) + (2i + 9)\big) = \sum_{i=0}^{2} (8i + 18) = (18) + (8 + 18) + (16 + 18) = 24 + 54 = 78$
  - $\sum_{m=3}^{5} \sum_{n=2}^{4} (m - n)(m + n)$
    - $\sum_{m=3}^{5} \sum_{n=2}^{4} (m - n)(m + n) = \sum_{m=3}^{5} \sum_{n=2}^{4} (m^2 - n^2) = \sum_{m=3}^{5} \big((m^2 - 2^2) + (m^2 - 3^2) + (m^2 - 4^2)\big) = \sum_{m=3}^{5} (3m^2 - 29) = (3 * 3^2 - 29) + (3 * 4^2 - 29) + (3 * 5^2 - 29) = 3 * 50 - 29 - 29 - 29 = 3 * (50 - 29) = 63$

Extension 1

- 1. Find a recurrence relation for the sequence: $a_n = 2^{1/2^n}$ (there could be many solutions)
  - $a_n = 2^{1/2^n} = 2^{1/(2^{n-1}\cdot 2)} = (2^{1/2^{n-1}})^{1/2} = \sqrt{a_{n-1}}$

Extension 2

- 2. The summation of a arithmetic progression

$$S = \sum_{i=1}^{n} i = 1 + 2 + \cdots + n \quad ①$$

$$= n + (n-1) + \cdots + 1 \quad ②$$

② + ① $2S = (n+1) + (n+1) + \cdots + (n+1) = n(n+1)$

- The summation of a geometric progression

$$S = \sum_{i=1}^{n} ar^i = ar + ar^2 + \cdots + ar^n \quad ③$$

$$\mathrm{r}S = \sum_{i=2}^{n+1} ar^i = ar^2 + ar^3 + \cdots + ar^n + ar^{n+1} \quad ④$$

④ - ③ $\mathrm{r}S - S = ar^{n+1} - a$